



Project
Overview

Project Features
& Sensors Used

Circuit Diagram
Implementation

Data Collection
& Processing

Workflow
Diagram

Demonstration

Future Work &
References

Wireless Sensor Network based Smart Healthcare Monitoring System

CS314 Wireless Sensor Network
Indian Institute of Information Technology, Vadodara



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Overview: Design and implement a wireless sensor network to enable continuous, real-time health and environment monitoring in hospitals.

- **Course Code and Name:** CS314 Wireless Sensor Network
- **Group Members:**
 - Isha Jangir (202211031)
 - Rajat Kumar Thakur (202211070)
- **References:** Kaidi, H. M., Izhar, M. A. M., Dziyauddin, R. A., Shaiful, N. E., & Ahmad, R. (2024). A Comprehensive Review on Wireless Healthcare Monitoring: System Components.



Project Features & Sensors

Key Features:

- Wearable physiological sensors (e.g., temperature, SpO₂)
- Environmental sensors (temperature & humidity)
- Wireless data transmission to the central dashboard
- Automated alert generation for abnormal readings

Sensor Type	Model	Purpose
Temperature	DS18B20	Accurate body or ambient temperature
Pulse Oximetry (SpO ₂ & HR)	MAX30100	Measures blood oxygen saturation & Heart Rate
Temperature & Humidity	DHT22	Monitors room conditions for patient comfort

Fig.: Sensors Used and Their Purpose



Circuit Diagram & Implementation

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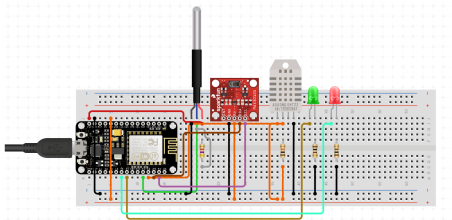


Fig.: Diagram created on Software

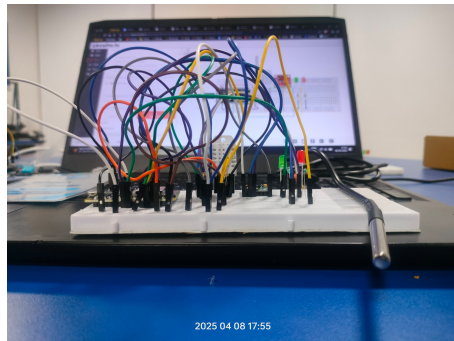


Fig.: Real life implementation



Data Collection & Processing

Data Collection and Processing:

Every two seconds, the NodeMCU reads your temperature, blood-oxygen, and room humidity. It sends the same data to a locally hosted web server. Finally, you open a simple web page on your phone or computer that shows live charts and widgets so you can monitor everything in real time.

Highlights:

- **Microcontroller:** NodeMCU (ESP8266) reads sensors via GPIO.
- **Local Logging:** Sends timestamped JSON data to the web server for redundancy and analysis.
- **Dashboard:** Uses AJAX to feed Chart.js charts and UI widgets; supports real-time and historical data views.



Workflow Diagram

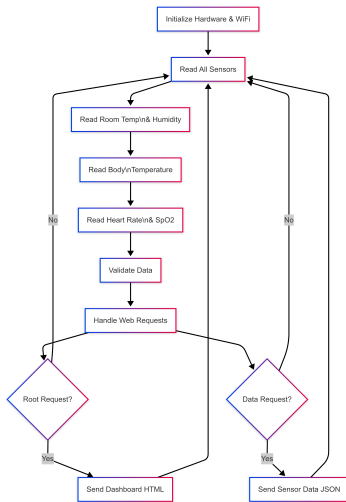


Fig.: Workflow Diagram



Demonstration

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- Circuit Diagram Implementation
- Data Collection & Processing
- Workflow Diagram
- Demonstration
- Future Work & References

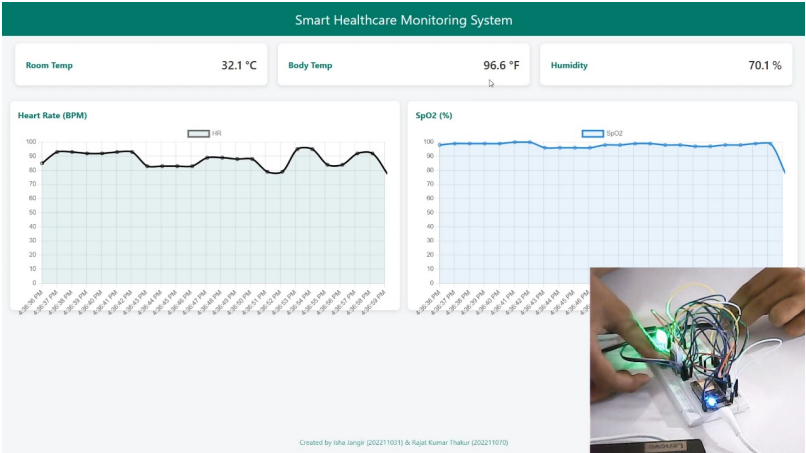


Fig.: Demonstration



Future Work & References

Future Work :

- **Anomaly Detection using Machine Learning:** Train models to detect abnormal patterns in vital signs and alert caregivers early.
- **Multi-Patient Monitoring Support:** Expand the system to handle and visualize data from multiple patients simultaneously.
- **Energy Efficiency & Battery Operation:** Use low-power modes, sleep cycles, and solar panels to extend battery life of sensor nodes.
- **Integration with Smart Home Systems:** Connect the system to IoT devices like smart beds or thermostats for complete environment control.

Additional References:

Ilyas, M. (2018). Wireless Sensor Networks for Smart Healthcare.